3.2.11 Elliptic Integrals and Elliptic Functions

`EllipticK[`*m*`]`	complete elliptic integral of the first kind $K(m)$
`EllipticF[`*phi*`, `*m*`]`	elliptic integral of the first kind $F(\phi\|m)$
`EllipticE[`*m*`]`	complete elliptic integral of the second kind $E(m)$
`EllipticE[`*phi*`, `*m*`]`	elliptic integral of the second kind $E(\phi\|m)$
`EllipticPi[`*n*`, `*m*`]`	complete elliptic integral of the third kind $\Pi(n\|m)$
`EllipticPi[`*n*`, `*phi*`, `*m*`]`	elliptic integral of the third kind $\Pi(n;\phi\|m)$
`JacobiAmplitude[`*u*`, `*m*`]`	amplitude function $\mathrm{am}(u\|m)$
`JacobiSN[`*u*`, `*m*`]`, `JacobiCN[`*u*`, `*m*`]`, etc.	Jacobi elliptic functions $\mathrm{sn}(u\|m)$, etc.
`InverseJacobiSN[`*v*`, `*m*`]`, `InverseJacobiCN[`*v*`, `*m*`]`, etc.	inverse Jacobi elliptic functions $\mathrm{sn}^{-1}(v\|m)$, etc.
`EllipticTheta[`*a*`, `*u*`, `*q*`]`	elliptic theta functions $\theta_a(u\|q)$ $(a = 1, ..., 4)$
`EllipticLog[{`*x*`, `*y*`}, {`*a*`, `*b*`}]`	generalized logarithm associated with the elliptic curve $y^2 = x^3 + ax^2 + bx$
`EllipticExp[`*u*`, {`*a*`, `*b*`}]`	generalized exponential associated with the elliptic curve $y^2 = x^3 + ax^2 + bx$
`ArithmeticGeometricMean[`*a*`, `*b*`]`	the arithmetic-geometric mean of a and b

Elliptic integrals and elliptic functions.

You should be very careful about the arguments you give to elliptic integrals and elliptic functions in *Mathematica*. There are several incompatible conventions in common mathematical use. You will often have to convert from a particular convention to the one that *Mathematica* uses.

In mathematical usage, the different argument conventions are sometimes distinguished by the use of separators other than commas between the arguments. Often, however, there is no clue about which notation is used, other than perhaps the specific names given to the arguments. In addition, in many cases, some arguments are not explicitly given.

Amplitude ϕ (used by *Mathematica*, in radians)

Delta amplitude $\Delta(\phi)$: $\Delta(\phi) = \sqrt{1 - m\sin^2(\phi)}$

Coordinate x: $x = \sin(\phi)$

Parameter m (used by *Mathematica*): preceded by $|$, as in $I(\phi|m)$

Complementary parameter m_1: $m_1 = 1 - m$

Modular angle α: preceded by $\backslash$, as in $I(\phi\backslash\alpha)$; $m = \sin^2(\alpha)$

Modulus k: preceded by comma, as in $I(\phi, k)$; $m = k^2$

Nome q: preceded by comma in θ functions; $q = \exp(-\pi K(1-m)/K(m))$

Characteristic n (used by *Mathematica* in elliptic integrals of the third kind)

Argument u (used by *Mathematica*): related to the amplitude by $\phi = \mathrm{am}(u)$

Invariants g_2, g_3 (used by *Mathematica*)

Periods ω, ω': $g_2 = 60\sum_{r,s}\frac{1}{w^4}$, $g_3 = 140\sum_{r,s}\frac{1}{w^6}$, where $w = 2r\omega + 2s\omega'$

Parameters of curve a, b (used by *Mathematica*)

Coordinate y (used by *Mathematica*): related by $y^2 = x^3 + ax^2 + bx$

Arguments for elliptic integrals and elliptic functions.

Elliptic Integrals

Integrals of the form $\int R(x, y)\, dx$, where R is a rational function, and y^2 is a cubic or quartic polynomial in x, are known as **elliptic integrals**. Any elliptic integral can be expressed in terms of the three standard kinds of **Legendre-Jacobi elliptic integrals**.

The **elliptic integral of the first kind** `EllipticF[`*phi*`, `*m*`]` is given by $F(\phi|m) = \int_0^\phi (1 - m\sin^2(\theta))^{-\frac{1}{2}}\, d\theta = \int_0^{\sin(\phi)} [(1-t^2)(1-mt^2)]^{-\frac{1}{2}}\, dt$. This elliptic integral arises in solving the equations of motion for a simple pendulum. It is sometimes known as an **incomplete elliptic integral of the first kind**.

Note that the arguments of the elliptic integrals are sometimes given in the opposite order from what is used in *Mathematica*.

The **complete elliptic integral of the first kind** `EllipticK[`*m*`]` is given by $K(m) = F(\frac{\pi}{2}|m)$. Note that K is used to denote the *complete* elliptic integral of the first kind, while F is used for its incomplete form. In many applications, the parameter m is not given explicitly, and $K(m)$ is denoted simply by K. The **complementary complete elliptic integral of the first kind** $K'(m)$ is given by $K(1-m)$. It is often denoted K'. K and iK' give the "real" and "imaginary" quarter-periods of the corresponding Jacobi elliptic functions discussed below. The **nome** q is given by $q(m) = \exp[-\pi K'(m)/K(m)]$.

The **elliptic integral of the second kind** `EllipticE[`phi`, `m`]` is given by $E(\phi|m) = \int_0^{\phi}(1-m\sin^2(\theta))^{\frac{1}{2}}\,d\theta = \int_0^{\sin(\phi)}(1-t^2)^{\frac{1}{2}}(1-mt^2)^{-\frac{1}{2}}\,dt$.

The **complete elliptic integral of the second kind** `EllipticE[`m`]` is given by $E(m) = E(\frac{\pi}{2}|m)$. It is often denoted E. The complementary form is $E'(m) = E(1-m)$.

The **Jacobi zeta function** is given in terms of the elliptic integrals E, F and K by $Z(\phi|m) = E(\phi|m) - E(m)F(\phi|m)/K(m)$.

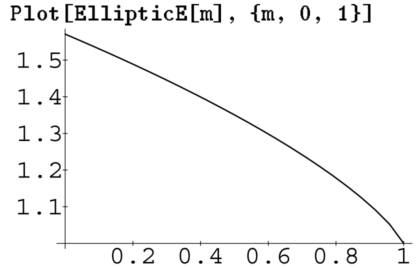
The **Heuman lambda function** is given by $\Lambda_0(\phi|m) = F(\phi|1-m)/K(1-m) + \frac{2}{\pi}K(m)Z(\phi|1-m)$.

The **elliptic integral of the third kind** `EllipticPi[`n`, `phi`, `m`]` is given by $\Pi(n;\phi|m) = \int_0^{\phi}(1-n\sin^2(\theta))^{-1}[1-\sin^2(\alpha)\sin^2(\theta)]^{-\frac{1}{2}}\,d\theta$.

The **complete elliptic integral of the third kind** `EllipticPi[`n`, `m`]` is given by $\Pi(n|m) = \Pi(n;\frac{\pi}{2}|m)$.

Here is a plot of the complete elliptic integral of the second kind $E(m)$.

```
In[1]:= Plot[EllipticE[m], {m, 0, 1}]
```

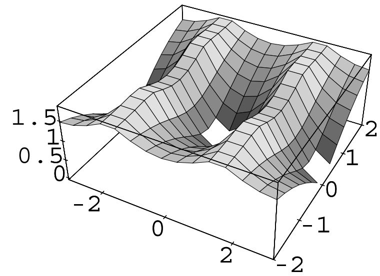


Here is $K(\alpha)$ with $\alpha = 30°$.

```
In[2]:= EllipticK[Sin[30 Degree]^2] // N
Out[2]= 1.68575
```

The elliptic integrals have a complicated structure in the complex plane.

```
In[3]:= Plot3D[Abs[EllipticF[px + I py, 1/2]],
                {px, -Pi, Pi}, {py, -2, 2}]
```



Elliptic Functions

Rational functions involving square roots of quadratic forms can be integrated in terms of inverse trigonometric functions. The trigonometric functions can thus be defined as inverses of the functions obtained from these integrals.

By analogy, **elliptic functions** are defined as inverses of the functions obtained from elliptic integrals.

The **amplitude** for Jacobi elliptic functions `JacobiAmplitude[`u`, `m`]` is the inverse of the elliptic integral of the first kind. If $u = F(\phi|m)$, then $\phi = \mathrm{am}(u|m)$. In working with Jacobi elliptic functions, the argument m is often dropped, so $\mathrm{am}(u|m)$ is written as $\mathrm{am}(u)$.

The **Jacobi elliptic functions** `JacobiSN[`u`, `m`]` and `JacobiCN[`u`, `m`]` are given respectively by $\mathrm{sn}(u) = \sin(\phi)$ and $\mathrm{cn}(u) = \cos(\phi)$, where $\phi = \mathrm{am}(u|m)$. In addition, `JacobiDN[`u`, `m`]` is given by $\mathrm{dn}(u) = \sqrt{1 - m\sin^2(\phi)} = \Delta(\phi)$.

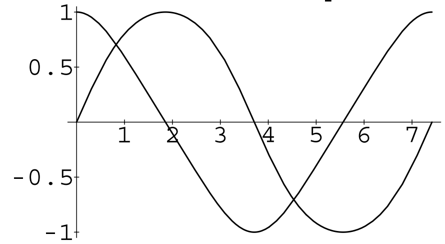
There are a total of twelve Jacobi elliptic functions `Jacobi`PQ`[`u`, `m`]`, with the letters P and Q chosen from the set `S`, `C`, `D` and `N`. Each Jacobi elliptic function `Jacobi`PQ`[`u`, `m`]` satisfies the relation $\mathrm{pq}(u) = \mathrm{pr}(u)/\mathrm{qr}(u)$, where for these purposes $\mathrm{pp}(u) = 1$.

There are many relations between the Jacobi elliptic functions, somewhat analogous to those between trigonometric functions.

The notation $\mathrm{Pq}(u)$ is often used for the integrals $\int_0^u \mathrm{pq}^2(t)\,dt$. These integrals can be expressed in terms of the Jacobi zeta function defined above.

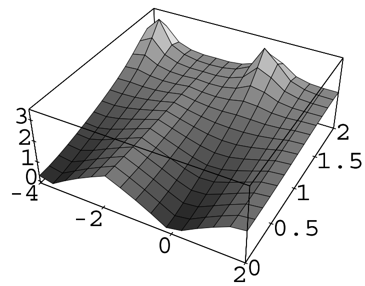
The first complete period of $\mathrm{sn}(u|\frac{1}{2})$ and $\mathrm{cn}(u|\frac{1}{2})$.

```
In[4]:= Plot[{JacobiSN[u, 1/2], JacobiCN[u, 1/2]},
             {u, 0, 4 EllipticK[1/2]}]
```



The Jacobi elliptic function $\mathrm{am}(u)$ has a complicated structure in the complex plane.

```
In[5]:= Plot3D[Abs[JacobiAmplitude[ux + I uy, 1/2]],
               {ux, -4, 2}, {uy, 0, 2}]
```



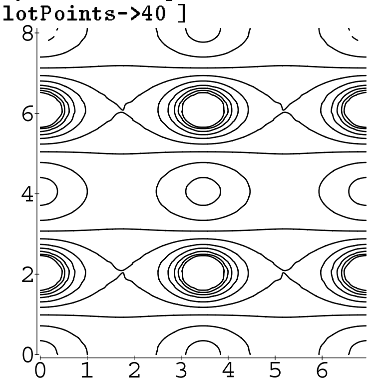
One of the most important properties of elliptic functions is that they are *doubly periodic* in the complex values of their arguments. Ordinary trigonometric functions are singly periodic, in the sense that $f(z + s\omega) = f(z)$ for any integer s. The elliptic functions are doubly periodic, so that $f(z + r\omega + s\omega') = f(z)$ for any pair of integers r and s.

The Jacobi elliptic functions $\mathrm{sn}(u|m)$ etc. are doubly periodic in the complex u plane. Their periods include $\omega = 4K(m)$ and $\omega' = 4iK(1 - m)$, where K is the complete elliptic integral of the first kind.

The choice of p and q in the notation $\mathrm{pq}(u|m)$ for Jacobi elliptic functions can be understood in terms of the values of the functions at the quarter periods K and iK'.

This shows one complete period in each direction of the Jacobi elliptic function $\mathrm{sn}(u|\frac{1}{3})$.

```
In[6]:= ContourPlot[Abs[JacobiSN[ux + I uy, 1/3]],
        {ux, 0, 4 EllipticK[1/3]},
        {uy, 0, 4 EllipticK[2/3]},
        PlotPoints->40 ]
```



Also built into *Mathematica* are the **inverse Jacobi elliptic functions** `InverseJacobiSN[`v`, `m`]`, `InverseJacobiCN[`v`, `m`]`, etc. The inverse function $\mathrm{sn}^{-1}(v|m)$ gives for example the value of u for which $v = \mathrm{sn}(u|m)$. The inverse Jacobi elliptic functions are related to elliptic integrals.

The four **elliptic theta functions** $\theta_a(u|m)$ are obtained from `EllipticTheta[`a`, `u`, `m`]` by taking a to be `1`, `2`, `3` or `4`. The functions are defined by: $\theta_1(u,q) = 2q^{\frac{1}{4}} \sum_{n=0}^{\infty} (-1)^n q^{n(n+1)} \sin((2n+1)u)$, $\theta_2(u,q) = 2q^{\frac{1}{4}} \sum_{n=0}^{\infty} q^{n(n+1)} \cos((2n+1)u)$, $\theta_3(u,q) = 1 + 2\sum_{n=1}^{\infty} q^{n^2} \cos(2nu)$, $\theta_4(u,q) = 1 + 2\sum_{n=1}^{\infty} (-1)^n q^{n^2} \cos(2nu)$. The theta functions are sometimes given in the form $\theta(u|m)$, where m is related to q by $q = \exp(-\pi K(1-m)/K(m))$. In addition, q is sometimes replaced by τ, given by $q = e^{i\pi\tau}$. All the theta functions satisfy a diffusion-like differential equation $\frac{\partial^2\theta(u,\tau)}{\partial u^2} = 4\pi i \frac{\partial\theta(u,\tau)}{\partial\tau}$.

The Jacobi elliptic function can be expressed as ratios of the theta functions.

An alternative notation for theta functions is $\Theta(u|m) = \theta_4(v|m)$, $\Theta_1(u|m) = \theta_3(v|m)$, $\mathrm{H}(u|m) = \theta_1(v)$, $\mathrm{H}_1(u|m) = \theta_2(v)$, where $v = \frac{\pi u}{2K(m)}$.

In terms of Θ and H, the **Neville theta functions** can be written $\theta_s(u) = \mathrm{H}(u)/\mathrm{H}_1(0)$, $\theta_c(u) = \mathrm{H}_1(u)/\mathrm{H}(u)$, $\theta_d(u) = \Theta_1(u)/\Theta(u)$, $\theta_n(u) = \Theta(u)/\Theta(0)$.

The **Weierstrass elliptic function** `WeierstrassP[`u`, `$g2$`, `$g3$`]` can be considered as the inverse of an elliptic integral. The Weierstrass function $\wp(u; g_2, g_3)$ gives the value of x for which $u = \int_{\infty}^{x} (4t^3 - g_2 t - g_3)^{-\frac{1}{2}}\, dt$. The function `WeierstrassPPrime[`$u$`, `$g2$`, `$g3$`]` is given by $\wp'(u; g_2, g_3) = \frac{\partial}{\partial u}\wp(u; g_2, g_3)$.

The Weierstrass functions are also sometimes written in terms of their *funda-*

mental periods ω and ω'.

Generalized Elliptic Integrals and Functions

The definitions for elliptic integrals and functions given above are based on traditional usage. For modern algebraic geometry, it is convenient to use slightly more general definitions.

The function `EllipticLog[{`x`, `y`}, {`a`, `b`}]` is defined as the value of the integral $\int_\infty^x (t^3 + at^2 + bt)^{-\frac{1}{2}}\, dt$, where the sign of the square root is specified by giving the value of y such that $y = \sqrt{x^3 + ax^2 + bx}$. Integrals of the form $\int_\infty^x (t^2 + at)^{-\frac{1}{2}}\, dt$ can be expressed in terms of the ordinary logarithm (and inverse trigonometric functions). You can think of `EllipticLog` as giving a generalization of this, where the polynomial under the square root is now of degree three.

The function `EllipticExp[`u`, {`a`, `b`}]` is the inverse of `EllipticLog`. It returns the list `{`x`, `y`}` that appears in `EllipticLog`. `EllipticExp` is an elliptic function, doubly periodic in the complex u plane.

`ArithmeticGeometricMean[`a`, `b`]` gives the **arithmetic-geometric mean** (**AGM**) of two numbers a and b. This quantity is central to many numerical algorithms for computing elliptic integrals and other functions. The AGM is obtained by starting with $a_0 = a$, $b_0 = b$, then iterating the transformation $a_{n+1} = \frac{1}{2}(a_n + b_n)$, $b_{n+1} = \sqrt{a_n b_n}$ until $a_n = b_n$ to the precision required.